

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. CHB-504: Physical Chemistry-III

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Why does a solvent flow spontaneously into a solution when separated by a semi-permeable membrane?
- (b) Differentiate between activity (a) and activity coefficient (γ).
- (c) Write the Debye-Hückel-Onsager equation for equivalent conductance.
- (d) What is the thickness (or radius) of the ionic cloud surrounding a reference ion?
- (e) What are the primary radiolytic products of water?

Question 2.

- (a) Discuss the concept and physical significance of activity and activity coefficient. Outline the conditions under which reference and standard states are defined for non-ideal solutions. [6]
- (b) State Raoult's law and Henry's law. Discuss the thermodynamic properties of ideal-dilute solutions. [8]

Question 3.

- (a) Give a qualitative idea of the Debye-Hückel theory of ion-ion interactions and show how these interactions are accounted for. [8]
- (b) Derive the relation between mean ionic activity coefficient ($\gamma_{\pm}$) and ionic strength (I) for dilute solutions. [6]

Question 4.

- (a) Discuss the different modes of interaction of γ -rays with matter (photoelectric effect, Compton scattering, and pair production). [6]
- (b) Discuss the primary radiolytic products and secondary chemical reactions in the radiolysis of an aqueous solution of ferrous sulphate (FeSO_4). [8]

Question 5. Write short notes on any two of the following:

[2 × 7 = 14]

- (a) Bravais lattices and crystal systems
- (b) Michaelis-Menten kinetics for enzyme catalysis
- (c) Nuclear fission and fusion processes